

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 50

Question Count: 5

Code of Conduct

I VAISHNAVI.S (192521313) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

A hypervisor is a virtualization software that allows multiple virtual machines (VMs) to run on a single physical server by managing resources such as CPU, memory, storage, and network. Hypervisors are mainly classified into two types: Type-1 hypervisor (Bare-Metal) and Type-2 hypervisor (Hosted). They differ in their architecture, performance, security, and management capabilities.

Type-1 Hypervisor

A Type-1 hypervisor runs directly on the physical hardware without requiring a host operating system. It directly controls the hardware resources and provides better performance and efficiency. It is mainly used in cloud datacenters and enterprise environments where high availability and reliability are required.

Examples: VMware ESXi, Microsoft Hyper-V, Xen.

Type-2 Hypervisor

A Type-2 hypervisor runs on top of an existing operating system such as Windows or Linux. It depends on the host operating system to access hardware resources. It is mainly used for testing, software development, and personal applications.

Examples: VMware Workstation, Oracle VirtualBox.

Comparison of Type-1 and Type-2 Hypervisors

1. Performance:

- Type-1 hypervisors provide higher performance because they directly access hardware resources.
- They have less overhead and provide faster execution of virtual machines.
- Type-2 hypervisors have lower performance because they depend on the host operating system.

2. Security:

- Type-1 hypervisors provide better security because they have fewer software layers and a smaller attack surface.
- Virtual machines are better isolated from each other.
- Type-2 hypervisors are less secure because vulnerabilities in the host operating system can affect all virtual machines.

3. Manageability:

- Type-1 hypervisors provide advanced management features such as live migration, resource allocation, backup, and high availability.
- They can manage thousands of virtual machines efficiently in cloud environments.
- Type-2 hypervisors provide basic management features and are suitable only for small-scale usage.

Analysis for Cloud Datacenter

A cloud datacenter hosting mission-critical applications requires high reliability, continuous availability, and strong security. Type-1 hypervisors are more suitable because they provide direct hardware access, efficient resource utilization, and better isolation between virtual machines. They also support features like load balancing and migration, which help maintain service availability.

Type-2 hypervisors are mainly designed for personal use and development environments. They are not preferred for mission-critical applications because the dependency on a host operating system reduces performance and increases security risks.

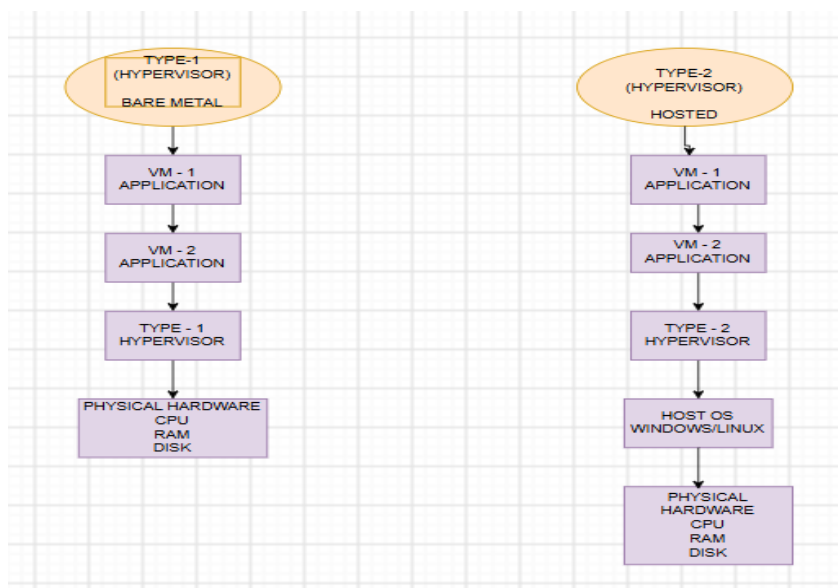
Justification

For hosting mission-critical applications in a cloud datacenter, Type-1 hypervisor is the better choice because:

- It provides better performance.
- It offers stronger security.
- It supports large-scale VM management.
- It improves resource utilization.
- It provides high availability and reliability.

Conclusion

Both Type-1 and Type-2 hypervisors are useful virtualization technologies, but their applications are different. Type-2 hypervisors are suitable for testing and small environments, whereas Type-1 hypervisors are ideal for cloud datacenters. Therefore, Type-1 hypervisor is preferred for mission-critical applications due to its superior performance, security, and manageability.



2.A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Virtualization architecture allows multiple virtual machines (VMs) to run independently on the same physical server using a hypervisor. Each virtual machine has its own operating system, applications, memory, storage, and network resources. Because of this separation, workloads from finance, healthcare, and student portals can safely share the same hardware without interfering with one another. In this case, the cloud provider hosts finance, healthcare, and student portal applications on the same physical server. These applications contain different types of sensitive information. Finance applications store banking and payment details, healthcare applications store patient records, and student portals contain academic and personal information. Therefore, proper isolation is necessary to ensure that one workload cannot access or interfere with another.

For example, if the finance application experiences heavy traffic or a software failure, it will not affect the healthcare or student portal virtual machines. Similarly, sensitive healthcare data remains isolated from other applications, helping organizations meet security and privacy requirements.

How Virtualization Supports Isolation:

- Each workload runs inside a separate virtual machine with its own operating system.
- The hypervisor allocates CPU, memory, storage, and network resources independently to each VM.
- Access between virtual machines is restricted unless explicitly permitted.
- A failure or malware attack in one VM normally does not spread to other VMs.
- Virtual networks and storage isolation further improve data security and workload separation.

Two Major Attack Surfaces:

1. Hypervisor Attack

The hypervisor is the core component managing all virtual machines. If an attacker compromises the hypervisor, they may gain control over multiple VMs running on the same physical server. Therefore, protecting and regularly updating the hypervisor is essential.

2. VM Escape Attack

In a VM escape attack, malicious software running inside one virtual machine exploits a vulnerability to escape its isolated environment and access the hypervisor or other virtual machines. This can lead to unauthorized access to sensitive data and services.

Conclusion:

Virtualization architecture provides strong isolation by keeping finance, healthcare, and student portal workloads in separate virtual machines while sharing the same physical hardware. This improves resource utilization, security, and reliability. However, cloud providers must protect the hypervisor and prevent VM escape attacks, as these are two of the most important security risks in a virtualized cloud environment.

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

A VM host utilization snapshot shows how efficiently the resources of a physical server are being used. In the given snapshot,

- CPU Utilization = 92%
- Memory Utilization = 88%
- Storage I/O Latency = 35 ms
- Average VM Boot Time = 170 seconds

These values indicate that the VM host is under heavy load. The CPU, memory and storage resources are almost fully utilized, which reduces the overall performance of the virtual machines.

1. CPU Bottleneck

The CPU utilization is 92%, which is very high. This means the processor is busy most of the time. Many virtual machines may be sharing the CPU, or some applications may be consuming more processing power.

Impact:

- Slow execution of applications.
- Delay in processing user requests.
- Reduced VM performance.

2. Memory Bottleneck

The memory utilization is 88%, which means most of the RAM is already occupied. When available memory becomes low, the system may start using virtual memory (disk), which is much slower than RAM.

Impact:

- Slow response from virtual machines.
- Delay in opening applications.
- Overall system performance decreases.

3. Storage Bottleneck

The storage I/O latency is 35 ms, which is higher than normal. This means the storage system is taking more time to read and write data. High latency usually occurs because many virtual machines are accessing the storage at the same time or the storage device is slow.

Impact:

- Slow file access.

- Delayed database operations.
- Slow loading of applications.

4. Slow VM Boot Time

The average VM boot time is 170 seconds, which is quite high. Normally, a virtual machine should boot much faster. The high boot time is mainly caused by the high CPU usage, high memory utilization and slow storage performance.

Impact:

- Virtual machines take longer to become available.
- Users experience delays.
- Cloud services become less efficient.

Recommended Corrective Actions

- Reduce CPU Load
 - Migrate some virtual machines to another host.
 - Upgrade the processor or increase CPU cores.
 - Stop unnecessary background processes.
- Increase Memory
 - Add more RAM to the host.
 - Allocate memory efficiently to each VM.
 - Remove idle or unused virtual machines.
- Balance the Workload
 - Distribute VMs across multiple physical hosts.
 - Use load balancing to avoid overloading a single server.
- Continuous Monitoring
 - Monitor CPU, memory and storage regularly.
 - Upgrade hardware whenever resource utilization remains consistently high.

Conclusion:

From the given utilization snapshot, it is clear that the VM host is facing CPU, memory and storage bottlenecks. The CPU utilization of 92%, memory utilization of 88%, storage I/O latency of 35 ms, and VM boot time of 170 seconds indicate that the server is overloaded. By balancing workloads, increasing CPU and memory resources, improving storage performance and monitoring the host regularly, the overall performance of the virtual machines can be significantly improved.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

A Software Defined Datacenter (SDDC) is a modern datacenter where all the resources such as compute, storage and networking are managed through software. In a traditional datacenter, most of these resources are configured manually, which takes more time and effort. SDDC improves agility by allowing administrators to manage the entire datacenter from a single software platform.

In a traditional datacenter, servers, storage and network devices are managed separately. Whenever a new application or virtual machine is required, the administrator has to manually configure the hardware, storage and network settings. This process is time-consuming and less flexible.

Limitations of Traditional Datacenter:

- Manual configuration takes more time.
- Difficult to scale resources quickly.
- Higher operational cost.

Software Defined Datacenter (SDDC)

In an SDDC, all the physical resources are virtualized and controlled using software. Resources can be allocated automatically according to the workload. This makes the datacenter more flexible, efficient and easier to manage.

How SDDC Improves Agility

Agility means the ability to respond quickly to changing business needs. SDDC improves agility in the following ways:

- New virtual machines can be created within minutes.
- Resources can be increased or decreased automatically.
- Applications can be deployed much faster.
- Management becomes easier through centralized control.
- Better utilization of hardware resources.
- Maintenance can be done with minimal downtime.

1. Compute Virtualization

Compute virtualization allows multiple virtual machines to run on a single physical server using a hypervisor.

Benefits:

- Better utilization of CPU and memory.
- Easy creation and migration of virtual machines.
- Quick allocation of computing resources.

- Reduced hardware cost.

Compared to a traditional datacenter, new servers can be deployed much faster without installing new physical hardware.

2. Storage Virtualization

Storage virtualization combines multiple storage devices into a single virtual storage pool. Administrators can allocate storage whenever required without changing the physical hardware.

Benefits:

- Easy storage management.
- Better storage utilization.
- Faster backup and recovery.
- Simple storage expansion.

3. Network Virtualization

Network virtualization creates virtual networks using software instead of depending only on physical switches and routers.

Benefits:

- Faster network configuration.
- Improved network security.
- Easy network isolation for different applications.
- Quick deployment of virtual networks.

Comparison Between Traditional Datacenter and SDDC:

conclusion:

Software Defined Datacenter (SDDC) provides greater agility than a traditional datacenter because it virtualizes compute, storage and network resources. It allows faster deployment, automatic resource allocation, easy scalability and centralized management. Compute virtualization improves server utilization, storage virtualization simplifies storage management, and network virtualization enables quick network configuration. Therefore, SDDC is more flexible, efficient and suitable for modern cloud computing environments.

5.A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

A multi-cloud federation allows organizations to use services from multiple cloud providers while maintaining secure access and resource sharing. In such an environment, identity management plays an important role because users need to access resources from different cloud platforms. The given problem occurs due to identity mismatches between cloud providers, which affects authentication and authorization.

Analysis of the Issue:

The failure occurs because different cloud providers may use different identity formats, authentication mechanisms, and access control methods. A user identity created in one cloud platform may not be recognized by another provider. This creates problems such as login failure, incorrect user permissions, difficulty in managing multiple accounts, and increased security risks. Without proper identity synchronization, unauthorized users may also gain access to sensitive cloud resources.

Secure Identity Design Using Federation Concepts:

To solve this issue, a Federated Identity Management System should be implemented. In this model, a trusted Identity Provider (IdP) manages user identities and authentication, while different cloud providers act as service providers. The cloud providers trust the identity information provided by the Identity Provider.

A Single Sign-On (SSO) mechanism can be used, allowing users to authenticate once and access multiple cloud services without maintaining separate accounts. Standard federation protocols such as SAML, OAuth 2.0, and OpenID Connect help in securely exchanging authentication information between different cloud providers.

Access control can be improved using Role-Based Access Control (RBAC), where permissions are assigned based on user roles. This ensures that users can access only the resources required for their work.

Privacy and Security Design:

To protect user privacy, only necessary identity information should be shared between cloud providers. Sensitive identity data should be encrypted during storage and transmission. Multi-Factor Authentication (MFA) should be implemented to provide additional security during login. Regular monitoring and audit logging should be used to track user activities and identify unauthorized access attempts.

Conclusion:

Identity mismatches between cloud providers can be resolved by implementing a secure federation-based identity architecture. Using a trusted Identity Provider, Single Sign-On, standard authentication protocols, and privacy protection methods ensures secure access management across multiple cloud environments. This improves interoperability, security, and user experience in a multi-cloud federation.